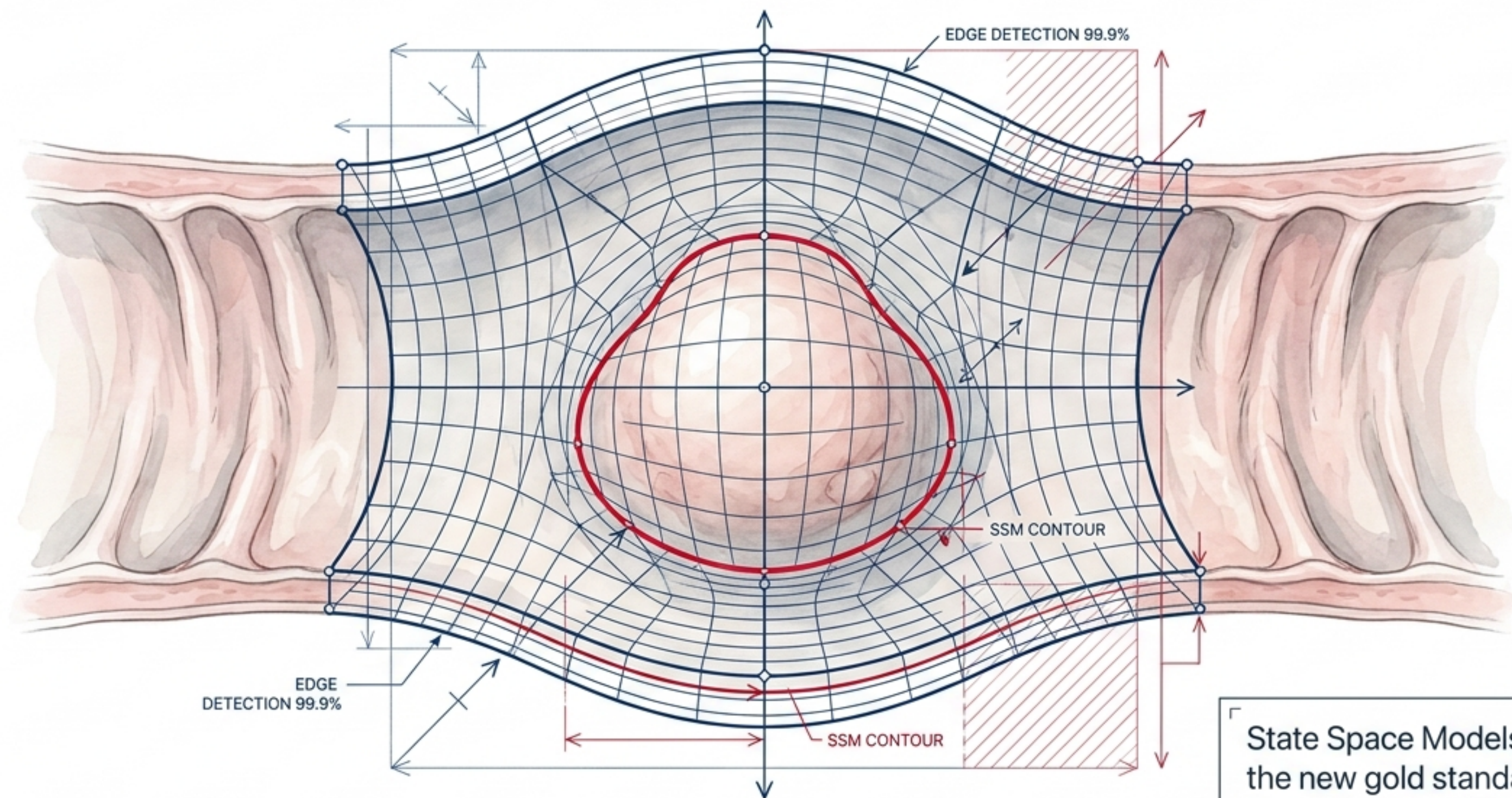


Decoding ϵ MMamba

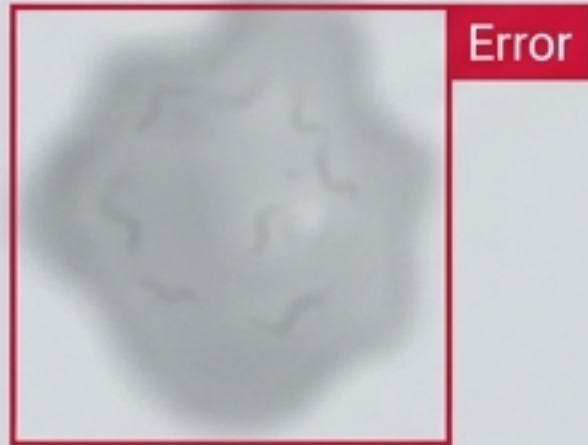


State Space Models (SSM) and the new gold standard for medical image segmentation.

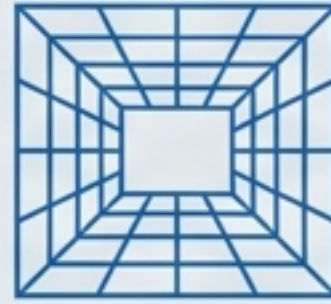
Standard AI fails in the hostile environment of the colonoscopy.



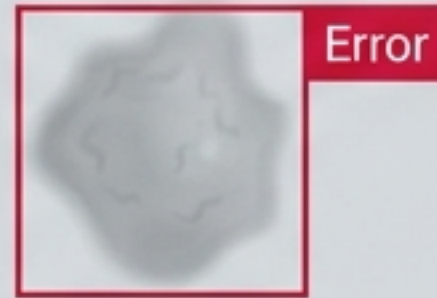
The Camouflage Problem



Polyps are made of the exact same mucosal tissue as the colon wall. They are pink bumps on a pink wall lacking high-contrast outlines.



The Scale & Context Problem



A colonoscope is constantly moving. A polyp might consume 80% of the frame, or appear as a tiny cluster of distant background pixels seconds later.

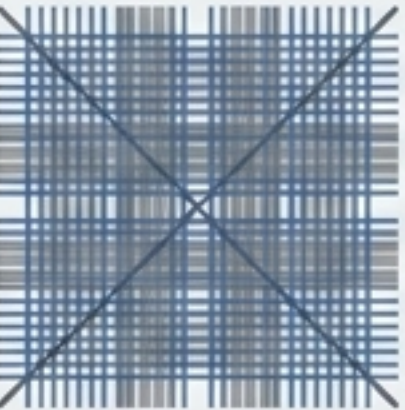


The Real-Time Speed Requirement

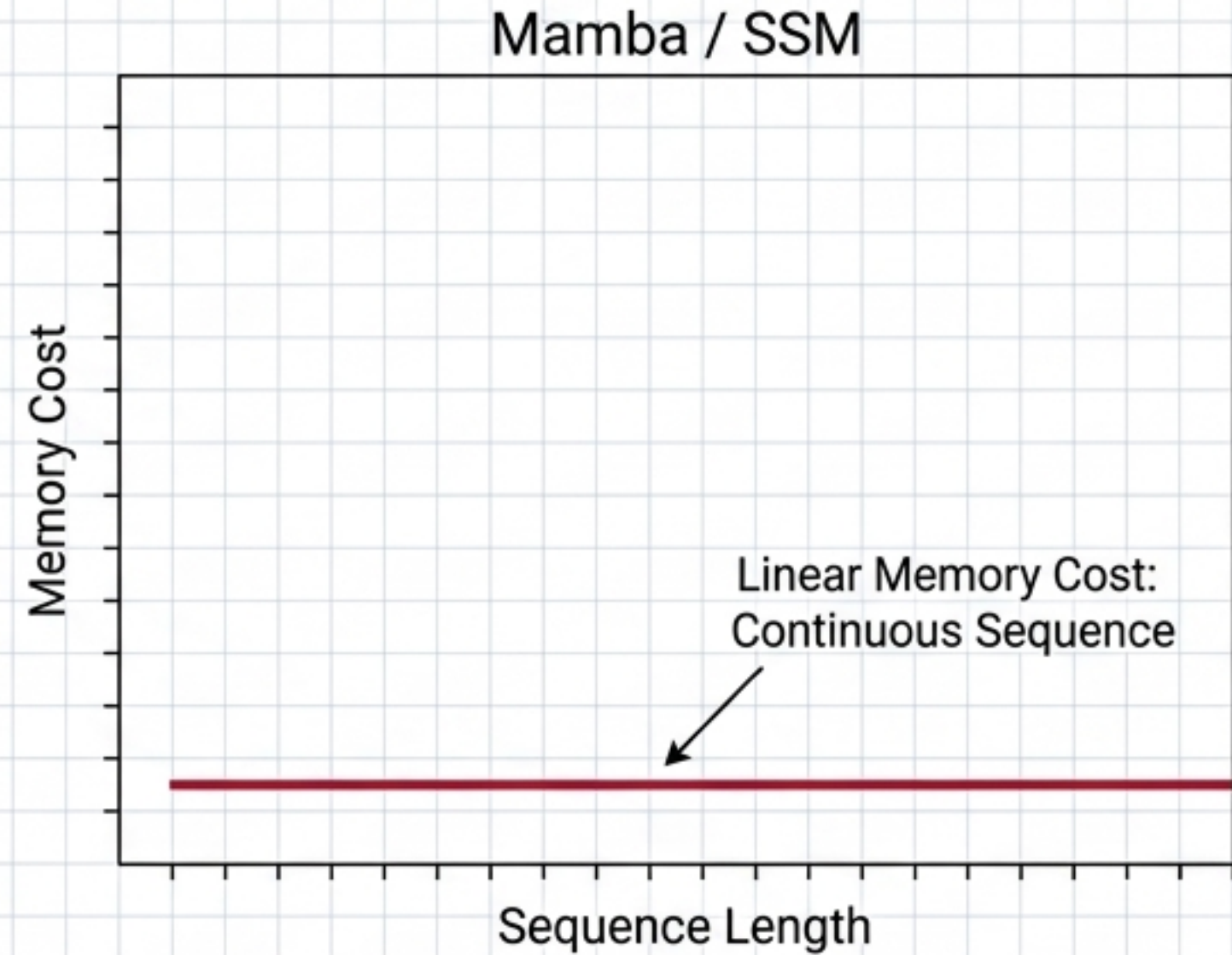
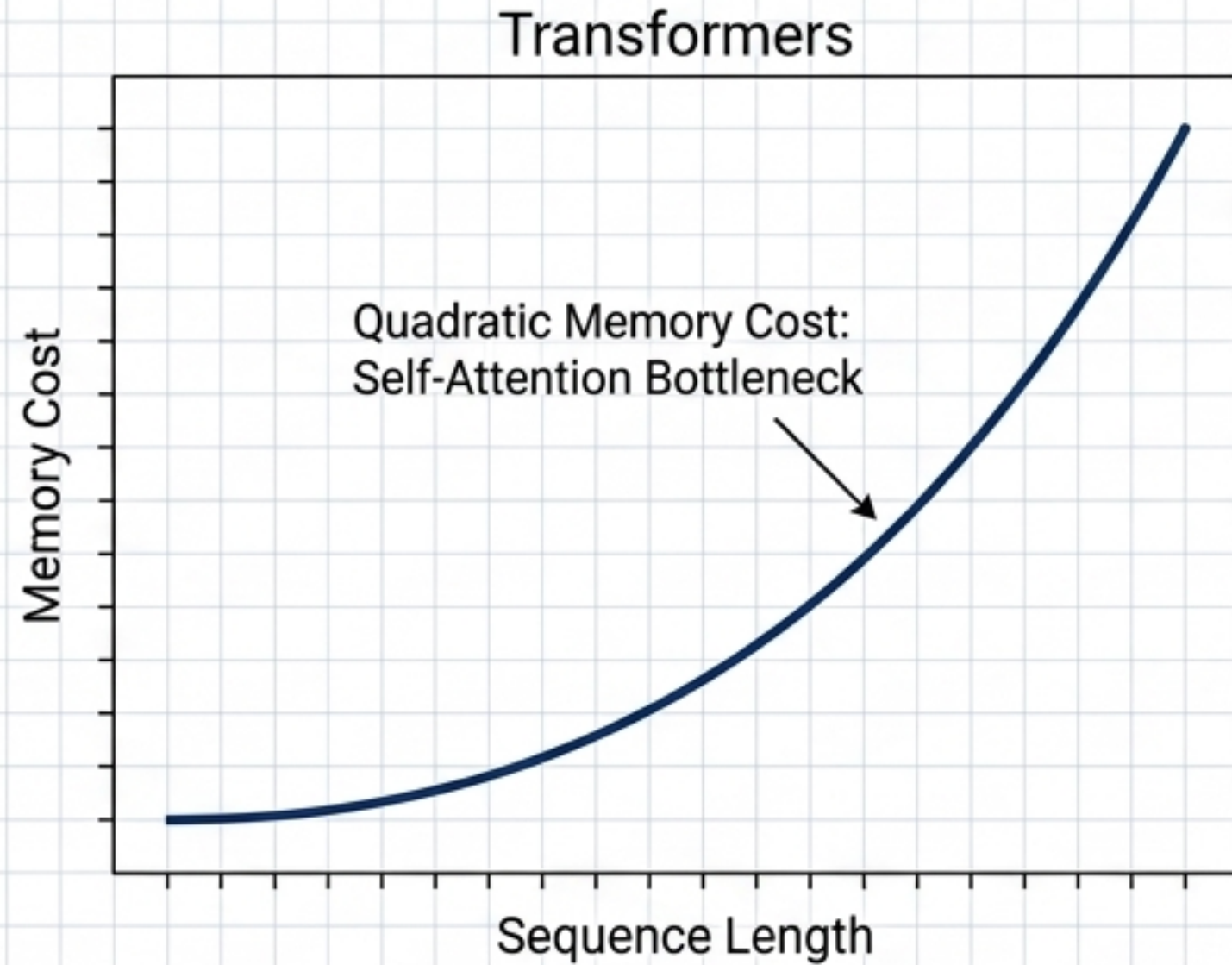


Diagnostics must happen live. Doctors require a minimum of 30 frames per second without relying on massive, expensive supercomputers in the OR.

The architectural roadblock in medical segmentation.

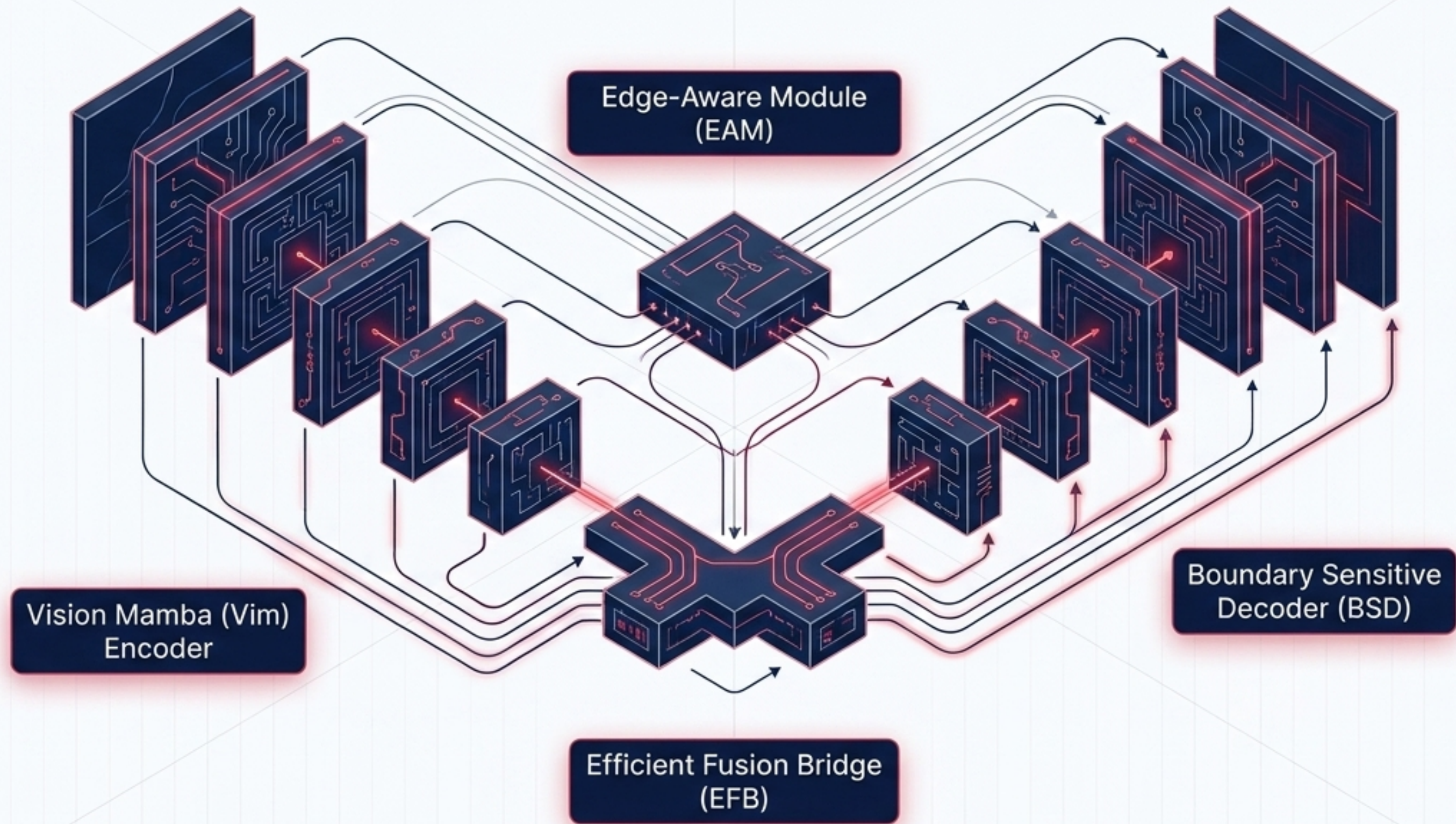
		Receptive Field	Computational Cost	Edge Precision
	Convolutional Neural Networks (CNNs)	Local. Constantly misses the broader context of the colon wall.	Linear (Fast).	Guessed. Results in jagged or shrunk masks around polyps.
	Vision Transformers (ViTs)	Global. Understands the whole scene.	Quadratic (Slow). Too heavy for real-time video without supercomputers.	Calculated, but computationally bottlenecked.
	ϵMMamba	Infinite & Global. Links tiny pixels to broader context instantly.	Linear (Fast). Achieves Transformer-level accuracy at real-time speeds.	Mathematically Forced. Perfectly traces near-invisible borders.

The Mamba paradigm breaks the speed-accuracy tradeoff.



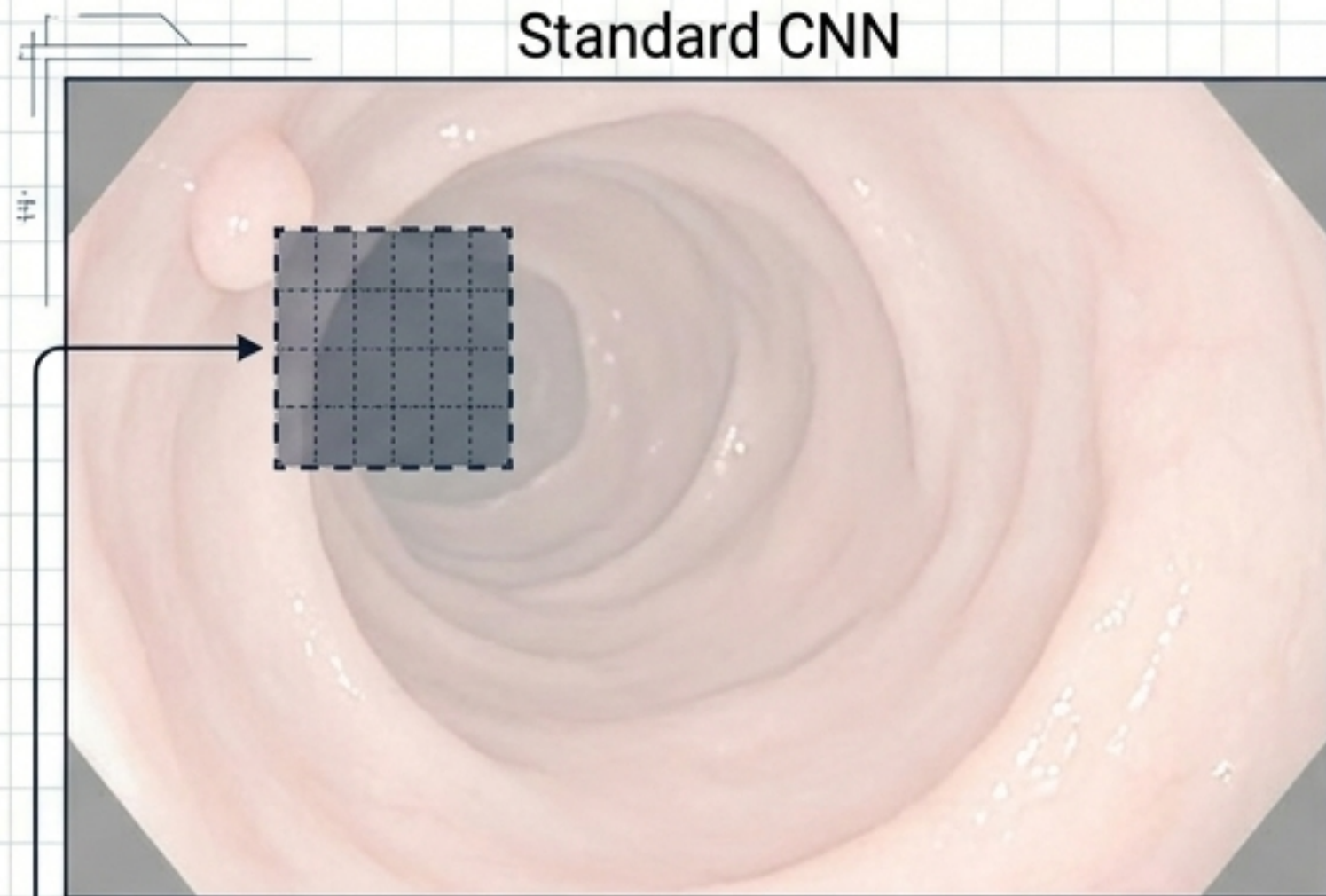
State Space Models (SSMs) provide an infinite, global receptive field but operate with the linear, highly efficient speed of a traditional convolution.

Inside the U-Net architecture of ϵ MMamba.



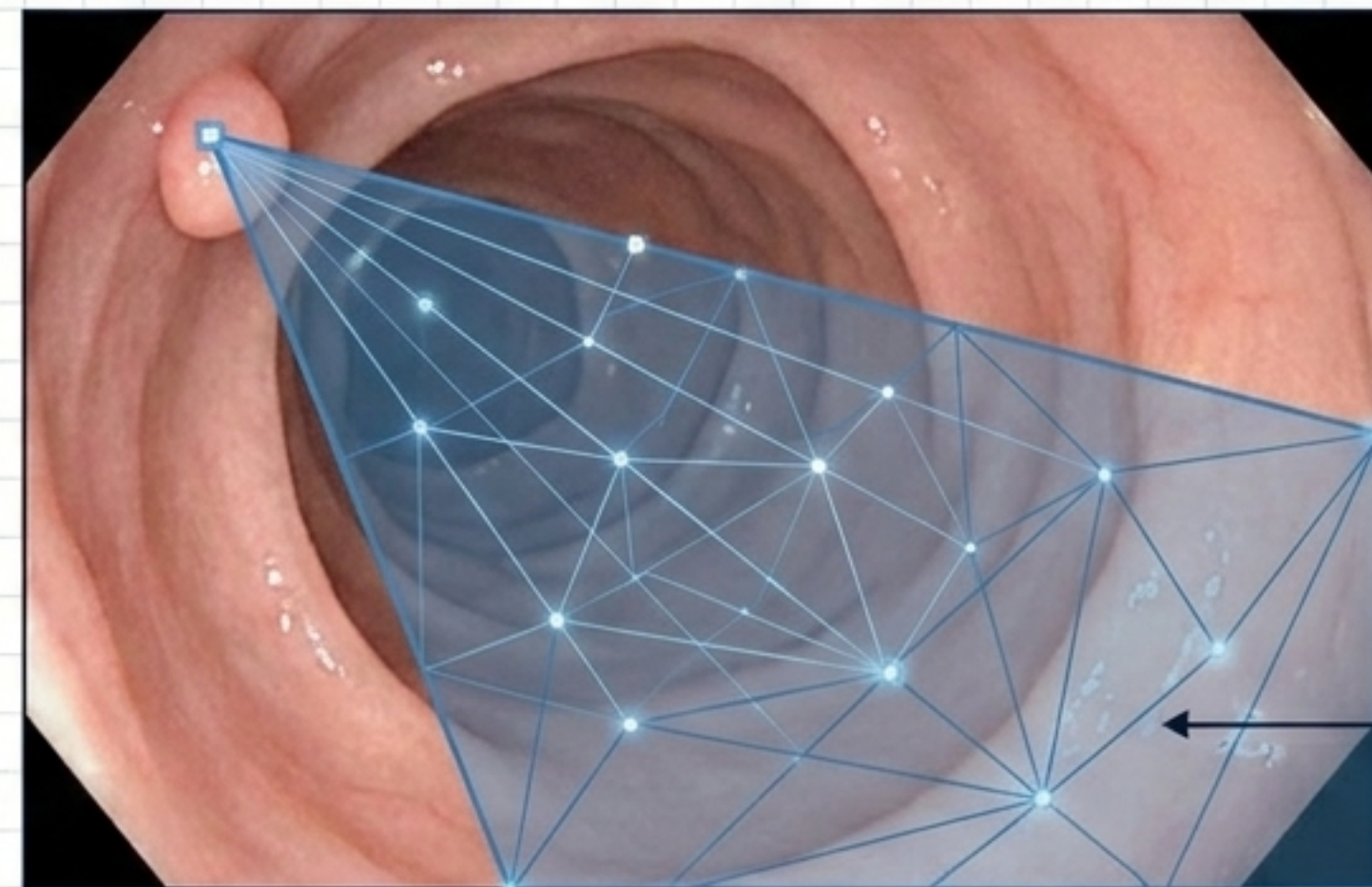
Capturing scale with the Vision Mamba (Vim) Encoder.

Standard CNN



Local patch processing.
Misses broader context.

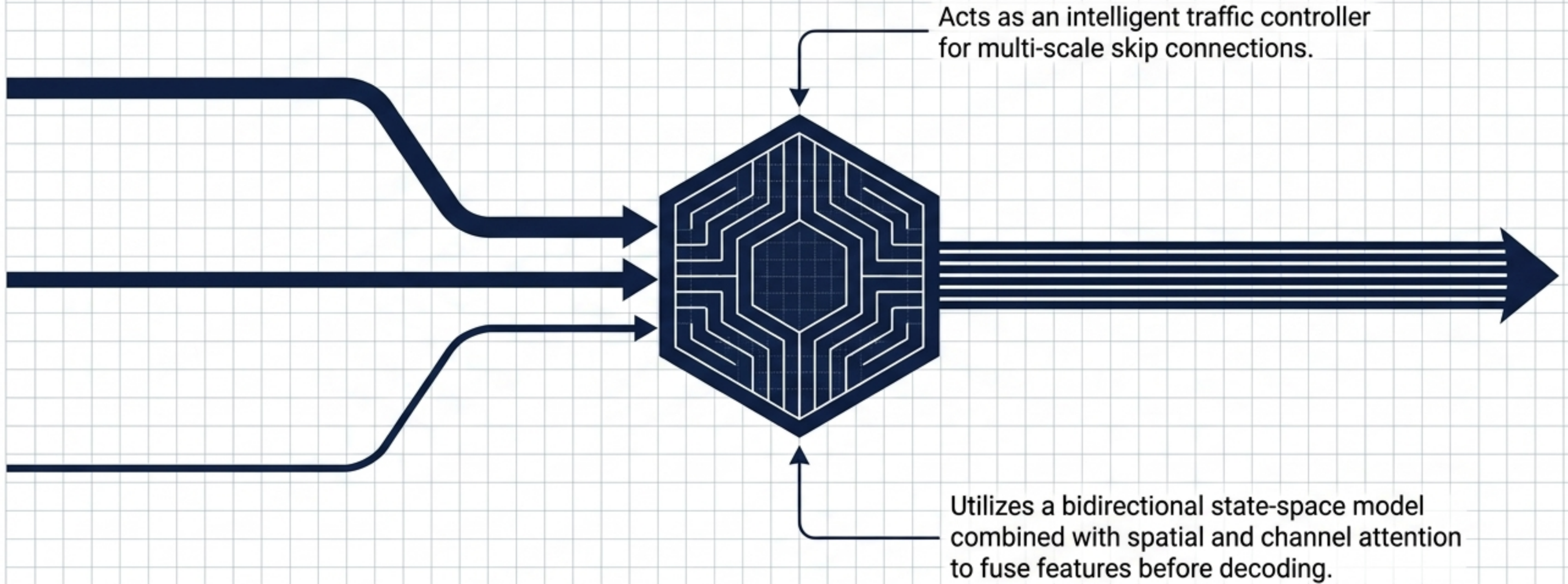
Vim Encoder



Global selective scan.
Mathematical link
established instantly.

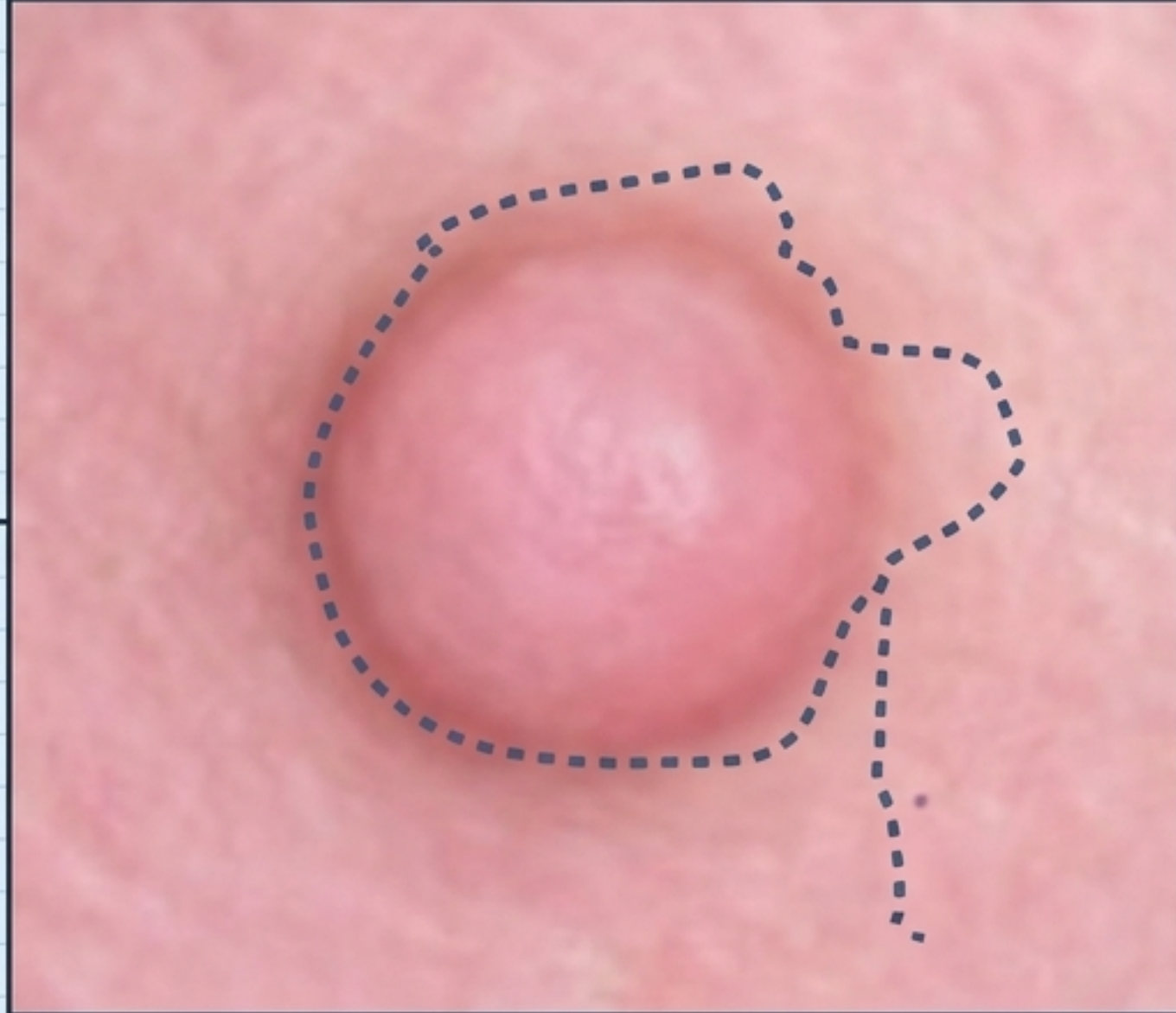
As the image passes through the network, Mamba blocks perform a selective scan over the pixels. It maps long-range dependencies in a single pass without the computational burden of self-attention.

Routing multi-scale features via the Efficient Fusion Bridge.



Eradicating camouflage with the Edge-Aware Module (EAM).

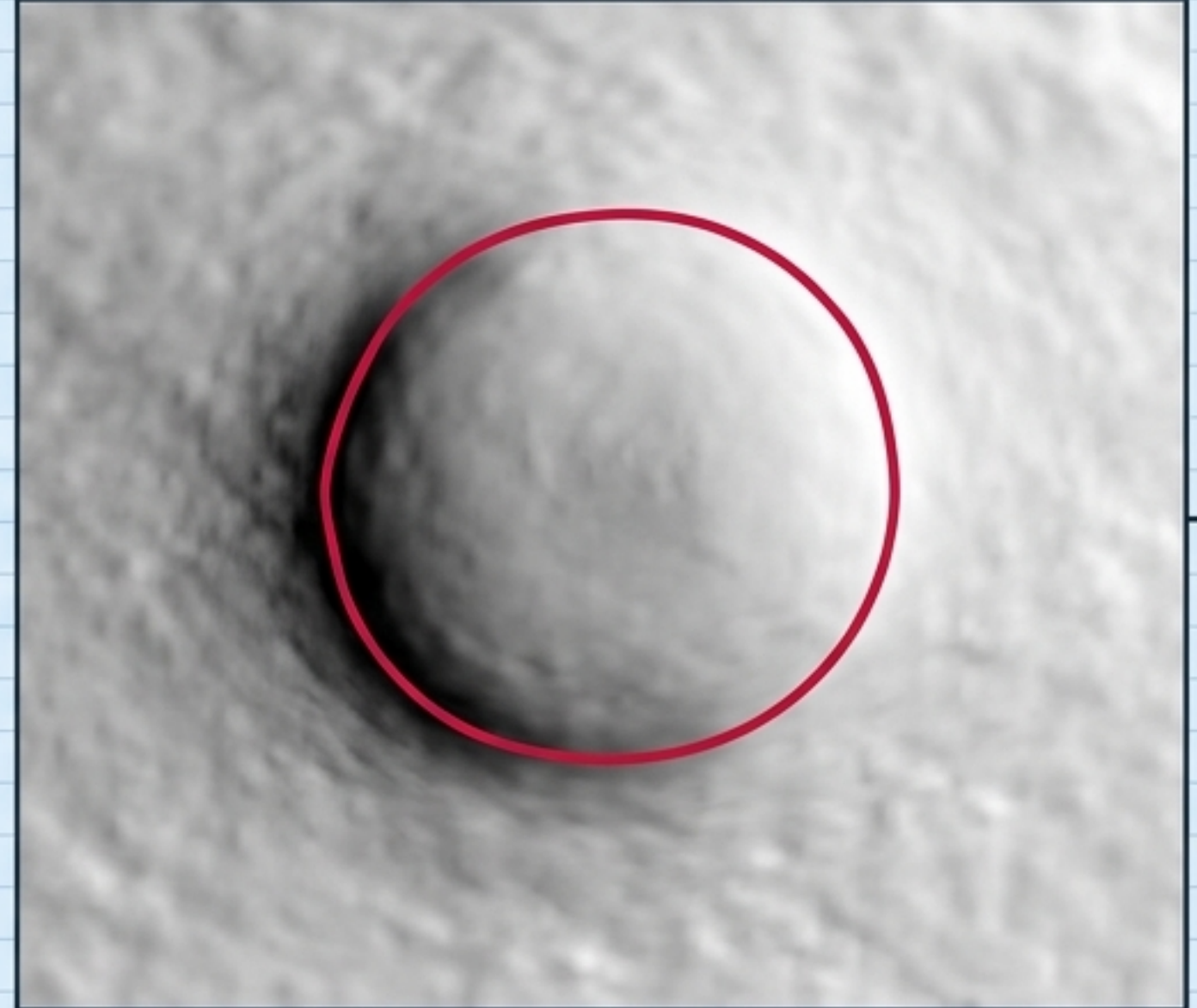
CNN Guesswork



Color Confusion: Ambiguous organic blending.



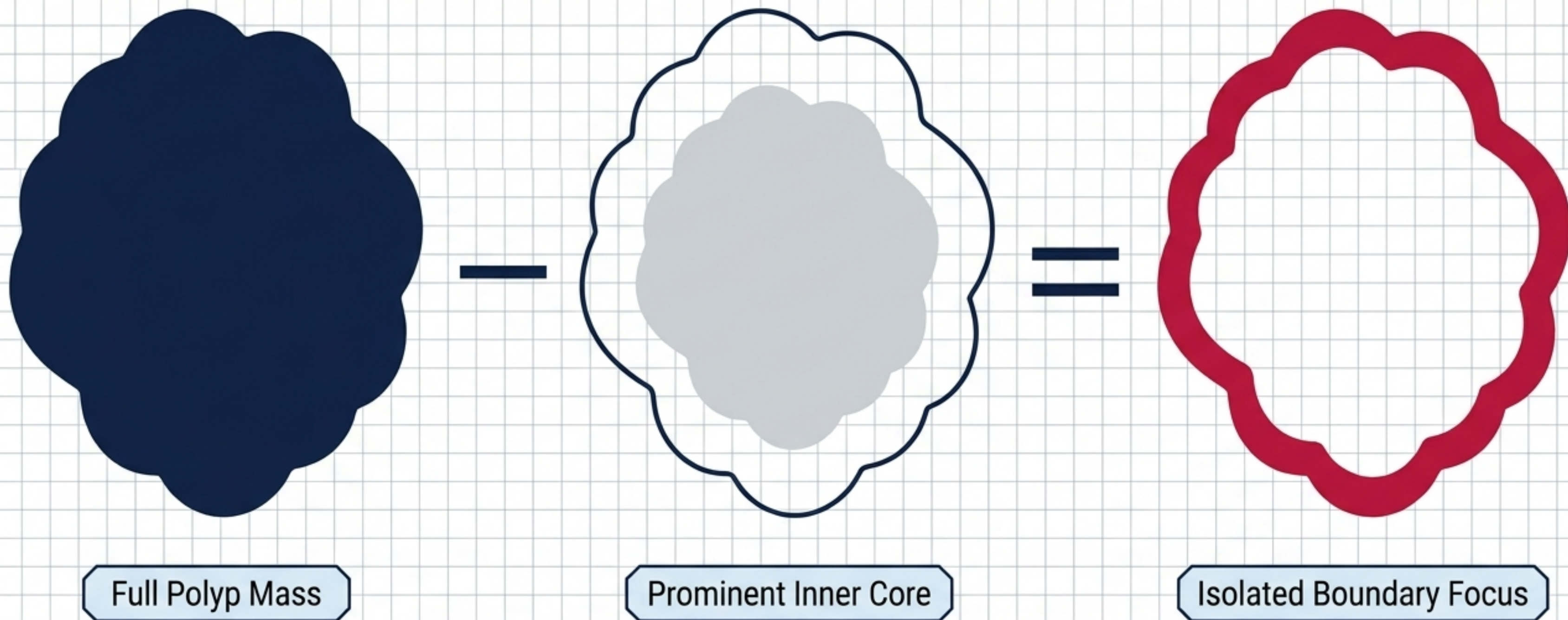
EAM Sobel Operator



Structural Clarity: Explicit mathematical gradients

Medical boundaries are blurry. Instead of guessing, the EAM explicitly calculates mathematical gradients using the Sobel edge detection algorithm. It forces the computer to see structural outlines over texture or color.

Isolating the perimeter with the Boundary Sensitive Decoder (BSD).



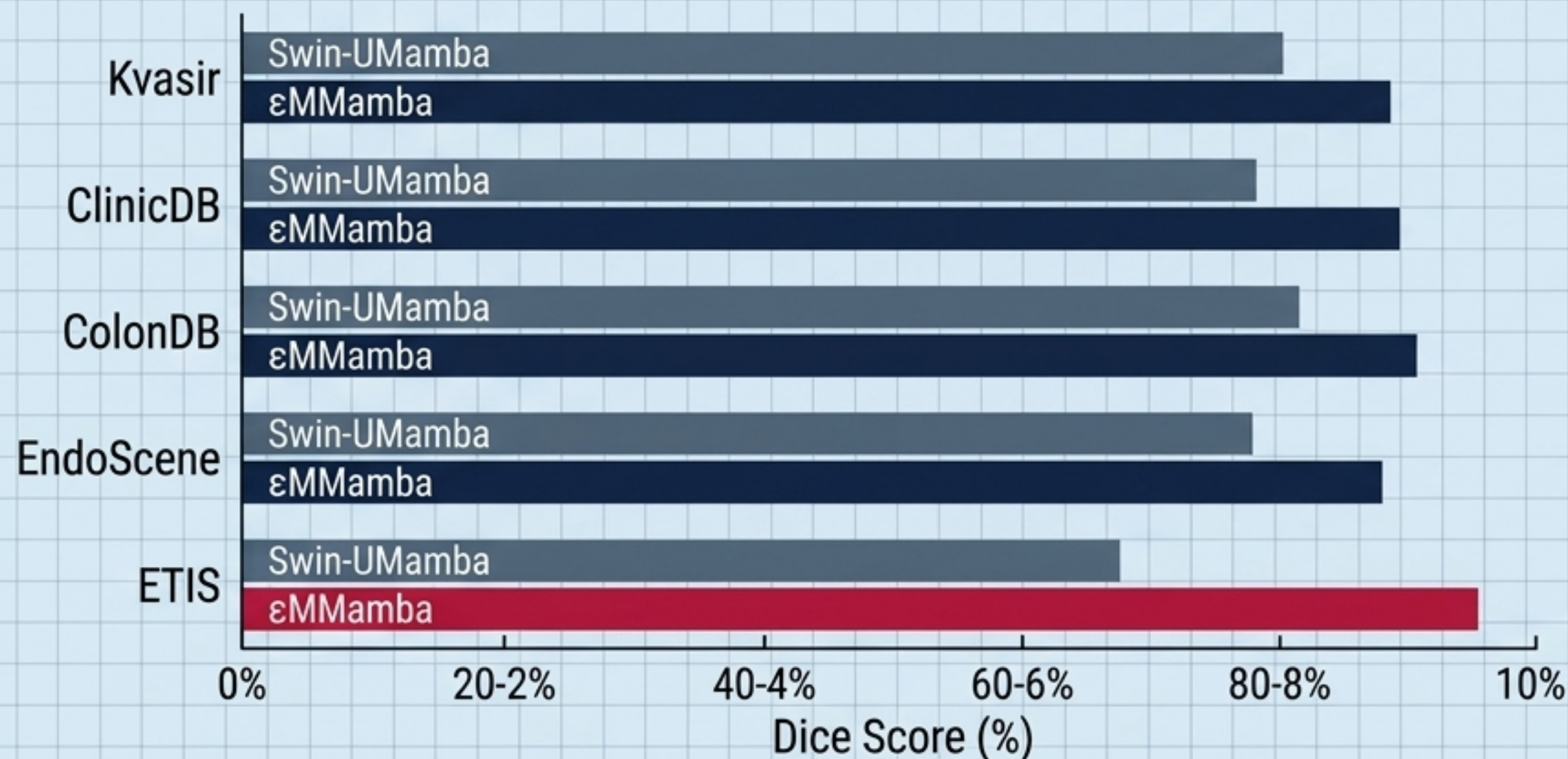
Standard attention looks at the easiest parts of an object. The BSD utilizes **inverse attention**. It mathematically subtracts the simple inner core, forcing the network to dedicate all processing power exclusively to tracing **difficult** boundary pixels.

Every line of code serves a clinical purpose.

Clinical Problem	Architectural Solution	Real-World Outcome
Low-Contrast Boundaries (Camouflage) 	Edge-Aware Module (Sobel gradients) + Boundary Sensitive Decoder (Inverse Attention)	Traces near-invisible mucosal borders with extreme precision.
Rapid Camera Movement (Scale/Context) 	Vision Mamba (Vim) Encoder (Infinite Receptive Field)	Seamlessly captures and adapts to polyps of any size in a single pass.
Live Feed Frame Rates (Speed) 	State Space Model Paradigm	Operates with linear time complexity for strict 30+ FPS hospital deployment.

The unprecedented benchmark of precision.

72.22 ETIS Dataset
Dice Score



εMMamba achieved an unprecedented Dice score of 72.22 on the ETIS dataset.

ETIS is the world's most challenging polyp dataset, filled with flat, small, and heavily camouflaged polyps.

A 72+ score definitively proves that explicit edge-enhancement methodology works in the most difficult clinical scenarios.